YUXUAN (VINCENT) ZHANG

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EDUCATION

Ph.D. in Physics, The University of Texas at Austin — Aug. 2017 – Aug. 2023
Advisors: Scott Aaronson and Andrew C. Potter
Specialized in **quantum information** and **many-body physics**
Thesis: *Exploring quantum matter in the era of quantum computers*

B.S. in Physics, University of California, Santa Barbara — Sep. 2013 – Jun. 2016
Graduated with Highest Honors (top 2.5% of the class)

EMPLOYMENT

Postdoctoral Fellow, EPFL & Princeton — Dec. 2025 – present
Supported by an ERC Consolidator Grant
Host: Dmitry Abanin

Research Scientist, BlueQubit Inc. — Dec. 2025 – present
Role: quantum algorithm development (part-time)

Postdoctoral Fellow, U of T & Vector Institute for Artificial Intelligence — Sep. 2023 – Dec. 2025
Host: Dvira Segal; (Quantum Software Consortium Grant) — Sep. 2025 – Dec. 2025
Hosts: Yong-Baek Kim; Juan Felipe Carrasquilla Álvarez; (CQIQC Fellow) — Sep. 2023 – Aug. 2025
Focus: synergy between **quantum information**, **quantum matter**, and **artificial intelligence**

PUBLICATIONS

Evaluating LLMs in Scientific Discovery — arXiv
Z. Song et al., including **Y. Zhang**
Featured by *Science news*
Under review, *Nat Comput Sci.*

Heuristic Quantum Advantage with Peaked Circuits — arXiv
H. Gharibyan, M. Z. Mullath, N. E. Sherman, V. P. Su, H. Tepanyan, and **Y. Zhang**

On verifiable quantum advantage with peaked circuit sampling — arXiv
S. Aaronson, **Y. Zhang** (corresponding author)
Under review, *Physical Review X*

Complexity and hardness of random peaked circuits — arXiv
Y. Zhang

Circuit compression for 2D quantum dynamics — arXiv
M. D'Anna, **Y. Zhang** (corresponding author), R. Wiersema, M. Rudolph, and J. Carrasquilla
Under review, *Science Advances*

Probing mixed-state phases on a quantum computer via Renyi correlators and variational decoding — arXiv
Y. Zhang, T. Hsieh, Y.B. Kim, and Y. Zou
Accepted, *Nature Communications*

Scalable quantum dynamics compilation via quantum machine learning — PRR
Y. Zhang, R. Wiersema, Y. B. Kim, J. Carrasquilla, and L. Cincio
Phys. Rev. Research 8, 023128

Biorthogonal Neural Network Approach to Two-Dimensional Non-Hermitian Systems PRL
M. Solinas, B. Barton, **Y. Zhang**, J. Nys, and J. Carrasquilla
Physical Review Letters, 136, 126501

Mixed-state phase transitions in spin-Holstein models PRB
B. Min, **Y. Zhang** (co-first author), Y. Guo, D. Segal, and Y. Ashida
Physical Review B, 111, 115123

Observation of a non-Hermitian supersonic mode on a trapped-ion quantum processor Nat Comm
Y. Zhang, J. Carrasquilla, and Y. B. Kim
Nature Communications, 16, 3286 (2025)
Featured in *Nature's Quantum Many-Body Dynamics Collection*

Classical Simulability of Circuits with Shallow Magic Depth PRX-Q
Y.F. Zhang, and **Y.X. Zhang** (corresponding author)
PRX Quantum, 6, 010337
Featured in the *PRX International Year of Quantum Collection*

Possibility of entanglement of purification to be less than half of the reflected entropy PRA
J. Couch, P. Nguyen, S. Racz, G. Stratis, **Y. Zhang** (corresponding author)
Physical Review A, 109, 022426

Sequential quantum simulation of spin chains with a single circuit QED device PRA
Y. Zhang, S. Shabani, A. Riswadkar, S. Shankar, and A. C. Potter
Physical Review A, 109, 022606

All-photonic one-way quantum repeaters npj-QI
D. Niu, **Y. Zhang**, A. Shabani, and H. Shapourian
npj Quantum Information 9, 106

Holographic Quantum Simulation of Entanglement Renormalization Circuits PRX-Q
S. Anand, J. Hauschild, **Y. Zhang**, A. C. Potter, and M. P. Zaletel
PRX Quantum 4, 030334

Quantum volume for photonic quantum processors PRL
Y. Zhang, D. Niu, A. Shabani, and H. Shapourian
Physical Review Letters, 130, 110602
CLEO conference 2023

Qubit-efficient simulation of thermal states with quantum tensor networks PRB
Y. Zhang, S. Jahanbani, D. Niu, R. Haghshenas, and A. C. Potter
Physical Review B, 106, 165126

Holographic simulation of correlated electrons on a trapped-ion quantum processor PRX-Q
D. Niu, R. Haghshenas, **Y. Zhang**, M. Foss-Feig, G. K. Chan, and A. C. Potter
PRX Quantum, 3, 030317

Straddling-gates problem in multipartite quantum systems PRA
Y. Zhang
Physical Review A, 105, 062430

QED driven QAOA for network-flow optimization Quantum
Y. Zhang, R. Zhang, and A. C. Potter
Quantum, 5, 510 (2021)

CEPC Conceptual Design Report Volume II: Physics & Detector arXiv
M. Abbrescia et al., including **Y. Zhang**

CEPC Conceptual Design Report: Volume 1 – Accelerator arXiv
M. Abbrescia et al., including **Y. Zhang**

PROFESSIONAL EXPERIENCE

Lecturer: School for Near Term Quantum Computing Benasque, Spain — Mar. 2026

Visitor Perimeter Institute for Theoretical Physics — Aug. 2023; Mar., May, Jul. 2024; May, Oct. 2025

Program Participant Kavli Institute for Theoretical Physics — Apr. 2024, Nov. 2025

Academic guest ETH Zurich — Nov. 2024 – Dec. 2024; Dec. 2025

Visitor Tsinghua University — Dec. 2023, Dec. 2024

Participant: Condensed Matter Summer School University of Minnesota — Jun. 2023

Graduate Research Assistant UT Austin — Jan. 2020 – Aug. 2023

Mentor: The Quantum Collective UT Austin — Sep. 2022 – May 2023

PhD Intern Cisco Systems, Inc. — May 2022 – Aug. 2022

Mentor: Directed Reading Program UT Austin — Jan. 2018 – May 2022

Chair and Representative Graduate Welfare Committee, UT Austin — Jun. 2020 – May 2022

Participant: Quantum Ideas Summer School Duke University — Jun. 2019

Teaching Assistant: Quantum Computing UT Austin — Jan. 2019 – Dec. 2019

Teaching Assistant: Physics Lab for Engineers UT Austin — Sep. 2017 – Dec. 2018

Visiting Researcher The Institute of High Energy Physics — Jul. 2016 – Aug. 2017

AWARDS AND FELLOWSHIPS

Vector Research Grant Vector Institute — Fall 2023 – Fall 2025

CQIQC Postdoctoral Fellowship UofT — Fall 2023 – Fall 2025

QuICS Hartree Postdoctoral Fellowship UMD (declined) — Spring 2023

Professional Development Award UT Austin — Spring 2023, Fall 2019

Lawrence C. Biedenharn Jr. Endowed Fellowship UT Austin — Fall 2017 – Spring 2018

Highest Honors at Graduation, UCSB — Summer 2016

Dean's Honors, UCSB — Fall 2013 – Spring 2016

PROFESSIONAL SERVICE

Below is a list of conferences and journals for which I have served as a reviewer, along with the initial invitation date:

Referee *Nature Communications* — Spring 2026

Referee *Quantum* — Fall 2025

Referee *npj Quantum Information* — Spring 2025

Referee *PRX Quantum* — Spring 2025

Referee *EPJ Quantum Technology* — Fall 2024

Referee *Physical Review A* — Fall 2024

Referee *Scientific Reports* — Summer 2024

Referee *Quantum Machine Intelligence* — Summer 2024

Referee *Theory of Quantum Computation, Communication, and Cryptography* Spring 2024

Referee *Physical Review Research* Fall 2023

Referee *Physical Review Letters* Summer 2023

Referee *Physical Review B* Spring 2023

Referee *Neural Networks* Spring 2023

Referee *International Symposium on Symbolic and Algebraic Computation* Fall 2023

Session Chair *APS March Meeting* Spring 2023

Abstract Sorter *APS March Meeting* Fall 2022

Referee *Quantum Information Processing* Spring 2021

STUDENT MENTORSHIP

Shuyi Liang, PhD at NUS (starting 2026) Research Paper, 2026

Brandon Barton, PhD at ETH Zürich Research Paper, 2026

Jakub Garwola, PhD at University of Toronto Research Paper, 2025

Xiang Zou, PhD at University of Toronto Research Paper, 2025

Michelle Ding, undergraduate at UT Austin→ PhD at UT Austin Honors Thesis, 2025

Xuheng Zhao, master at ETH Zürich Master Project, 2025

Ruize Ma, master at ETH Zürich → PhD at EPFL Research Paper, 2025

Matteo D'Anna, PhD at ETH Zürich Research Paper, 2025

Massimo Solinas, master at ETH Zürich → PhD at Regensburg University Research Paper, 2025

Yifan Zhang, PhD at Princeton Research Paper, 2024

Adam Martinez, undergraduate at UofT → PhD at Oxford Honors Thesis, 2024

Xiaoxiao (Alice) Xiong, undergraduate at UBC → Caltech Honors Thesis, 2023

Michelle Gelman, undergraduate at UT Austin → PhD at USC Honors Thesis, 2023

Shahin Jahanbani, undergraduate at UT Austin → PhD at UCB Research Paper, 2022

PRESENTATIONS AND INVITED TALKS

- "Verifiable Quantum Advantage: Why I'm Obsessed with Peaks," ETH Zurich, Zurich, Spring 2026
- "Toward verifiable and practical quantum advantage," NUS, Singapore, Spring 2026
- "Verifiable Quantum Advantage: Why I'm Obsessed with Peaks," NTU, Singapore, Spring 2026
- "Verifiable Quantum Advantage: Why I'm Obsessed with Peaks," Princeton University, Princeton, Spring 2026
- "A Physicist's Guide to Quantum Complexity and Quantum Supremacy," Spring School on Near-Term Quantum Computing, Benasque, Spring 2026
- "Verifiable quantum advantage with peaked circuit sampling," Tsinghua University, Beijing, Winter 2025
- "Classical Simulability of Quantum Circuits with Shallow Magic Depth," Year of Quantum Across Canada, Waterloo, Fall 2025

- "Co-design Research on Quantum Information Theory, Artificial Intelligence, and Quantum Chips," HKUST(GZ), Guangzhou, Summer 2025
- "Artificial Intelligence & Quantum: empowering the future of computation", Royal Bank of Canada, Halifax, Summer 2025
- "Compressing quantum dynamics with quantum machine learning," Fields Institute, Toronto, Summer 2025
- "Observation of a non-Hermitian supersonic mode on a trapped ion quantum computer," Tsinghua University, Beijing, Winter 2024
- "Observation of a non-Hermitian supersonic mode on a trapped ion quantum computer," University of California, Berkeley, Online, Winter 2024
- "Variational quantum compilation for dynamical simulation," University of California, Los Angeles, Online, Winter 2024
- "Variational quantum compilation for dynamical simulation," Fudan University, Shanghai, Winter 2024
- "Variational quantum compilation for dynamical simulation," Shanghai Jiao Tong University, Shanghai, Winter 2024
- "Observation of a non-Hermitian supersonic mode on a trapped ion quantum computer," ETH Zurich, Zurich, Fall 2024
- "Scalable quantum dynamics compilation via quantum machine learning," National University of Singapore, Online, Fall 2024
- "Scalable quantum dynamics compilation via quantum machine learning," Vector Institute, Toronto, Fall 2024
- "Observation of a non-Hermitian supersonic mode on a trapped ion quantum computer," University of Toronto, Toronto, Fall 2024
- "Advancing science in the era of quantum and AI", IBM-CQIQC meeting, Toronto, Spring 2024
- "Qubit-efficient quantum simulation with sequential circuits," Tsinghua University, Beijing, Winter 2023
- "Quantum volume for measurement-based quantum processors," APS March Meeting, Las Vegas, Spring 2023
- "Quantum volume for photonic quantum computing," Cisco Research, San Jose, Fall 2022
- "Holographic simulation of correlated electrons and thermal states on a trapped-ion quantum processor," APS March Meeting, Chicago, Spring 2022
- "Holographic simulation of correlated electrons and thermal states on a trapped-ion quantum processor," Brookhaven National Laboratory, Spring 2022
- "Interacting fermions on a quantum processor," Quantum Circuits, Inc., New Haven, Spring 2022
- "QED driven QAOA for network-flow optimization," Quantum Information Processing (QIP), Shenzhen, Winter 2020
- "Quantum computing today," The Institute of High Energy Physics, Beijing, Summer 2019